

PC protection

If you face constant voltage fluctuation and load shedding, read on for a better understanding of UPS systems

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I've always emphasised the importance of protection for desktop computers. Irrespective of how much you spend on your components, danger lurks in the wires that supply electricity to your home and your PC. A UPS is a device that provides uninterrupted and regulated power to sensitive devices such as a computer. All UPSs will allow your PC to run for a few extra minutes after a power cut allowing you to save your work and shut it down properly.

I've come across people with dead components all because of a failure to install a UPS. If the power supply to your PC suddenly gets cut, several things can happen. Nine times out of ten nothing happens, but there is a possibility of damage to your hard drive and RAM – usually in that order. The motherboard is

also at risk. At the very least, if nothing gets damaged you lose all unsaved data. A UPS eliminates these risks.

The first type of UPS is the "Stand By" type. This UPS simply switches the load to battery in case of power loss on the AC power line. The response time of the switchover (called switching time) is between 2-10 ms. Since most SMPSs have a hold-up time of no less than 16 ms, i.e. greater than the switching time of the UPS, there is no problem of shutdown between the transition from mains power to the DC to AC inverter on the UPS.

The most prevalent type of UPS today is the "line interactive UPS". These can regulate the input AC power to some extent and the battery is charged via AC power. When the AC power cuts, AC input is disconnected and the battery provides power. Such

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UPSs are also compact and as such this is what you will find available with most PC vendors. 99 per cent of PC users will want to purchase one of these and they provide adequate protection from surges and spikes and of course blackouts.

There is another type of UPS called the "online UPS". These are much costlier. In this design the AC mains charges only the battery via the inverter and therefore the output runs off only the battery. Since the AC line is never directly powering the connected equipment there is virtually no chance of damage through faults in the line, also there is no switching time involved.

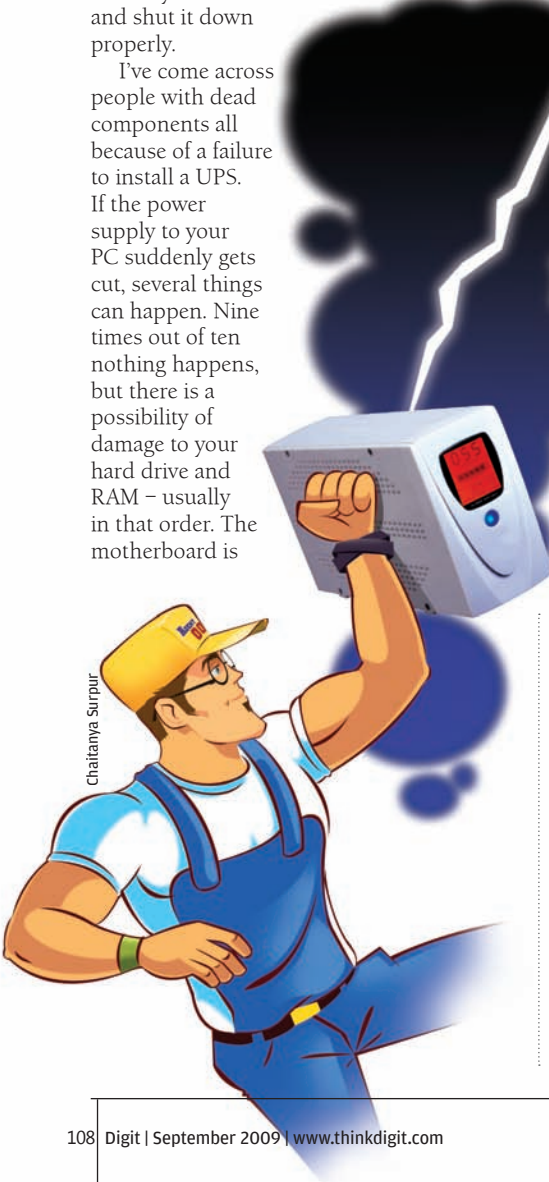
In my opinion when it comes to UPSs, do be choosy about brands. Remember the number of service centres and support available is secondary; the idea is to protect other components, meaning your UPS should not fail at any cost. I trust the following brands – APC, Numeric and PowerSafe. There are other brands such as

Datex, Microtek and Powercom that are also reliable but the first three should get a first look when purchasing. Personally I have owned a Numeric UPS for the last five years and am a very satisfied customer – not a single hiccup despite some gruelling use.

When deciding on the capacity of the UPS remember to use a conversion factor of 0.7. Therefore a UPS with a rating of 1000 VA (volt amperes) equates to a maximum output of 700 watts. You could either be clever and match this to the approximate power consumption of your PC or be really smart and match it to the output of your SMPS unit – either way works. Though the latter may be a bit costlier, it involves minimal calculations. Most people will sell you a UPS on the basis of its backup time – this is totally wrong as the battery backup totally depends on the amount of power your system draws – the more powerful the system, the quicker the battery will drain. Backup time should be used wisely to save and shut down all running programs and your PC, not to continue your work and hope the battery lasts. For most PCs, a UPS rated at 600 VA will suffice. If you are running a powerful PC with a quad core CPU and a fast graphics card you may want to look at something a little higher – 800VA should be good enough.

For most systems, APC's Back-UPS ES 650VA priced at Rs. 3,100 should suffice. If you want something a little beefier I swear by the Numeric Digital 800VA – this model is a little older, priced at Rs. 3,200 but offers sterling stability – this is the one I've been using for the last five years with no problems whatsoever. Alternatively, you could go for APC's BR800-IN priced at Rs. 4,700. If you want something beefier to protect up to two PCs, then the APC Back-UPS RS 1500VA is available for Rs. 8,900 although unfortunately it has only three 6 amp outputs for the battery backup which means you will need to invest in a power strip (spike guard).

A select few people will be looking for an online UPS, particularly those that have invested a lot in their PCs and wish to leave nothing to chance. I recommend the PowerSafe 1000ONL – an online UPS rated at 1000 VA and I was surprised to see it priced at just Rs. 10,000 – cheap for an online UPS. **d**



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